

LEVEL-I (MAIN)

EXERCISE-I

Straight Objective Type Questions

(IUPAC, Wernes theory, Isomerism)

- A bidentate ligand is
1) pyridine 2) thiocyanate 3) ethylene diammine 4) water
- The IUPAC name of the complex $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ is
1) pentaamminenitrito-N-cobalt(III) chloride 2) nitrito-N-pentaamminecobalt(III) chloride
3) nitrito-N-pentaamminecobalt(II) chloride 4) pentaamminenitrito-N-cobalt(II) chloride
- Which of the following is wrong with respect to $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
1) central metal ion is Co and the ligands are NH_3 and Cl^- ion
2) oxidation number of Co is +2
3) co-ordination number of Co is 6
4) the number of ions formed when 1 mole of the compound dissolves in water is 3 moles
- Which of the following is an example of ambidentate ligand
1) CO 2) CN^- 3) H_2O 4) SO_4^{2-}
- (A) : IUPAC name of $[\text{Pt Br}_2(\text{en})_2]\text{Cl}_2$ is dibromobis (1, 2-ethane diamine) platinum (IV) chloride.
(R) : (en) represents ethylenediamine and it is a ligand where prefix di- is already included in its designation.
1) Both (A) and (R) are true and (R) is the correct explanation of (A)
2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
3) (A) is true but (R) is false 4) (A) is false but (R) is true
- Neutral complex among the following
1) $\text{CuSO}_4 \cdot 4\text{NH}_3$ 2) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ 3) $\text{Ni}(\text{CO})_4$ 4) $[\text{Pt}(\text{NH}_3)_2]\text{Cl}_2$

7. The neutral complex, diamminedibromodichloroplatinum(IV) is best represented as
 1) $[\text{Pt}(\text{NH}_3)_2\text{Br}_2\text{Cl}_2]$ 2) $[\text{PtCl}_2\text{Br}_2(\text{NH}_3)_2]$ 3) $[\text{PtBr}_2\text{Cl}_2(\text{NH}_3)_2]$ 4) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2\text{Br}_2]$
8. Which of the following is an example of chelating ligand
 1) en 2) DMG 3) $\text{CH}_3\text{CH}_2\text{NH}_2$ 4) both (1) and (2)
9. The coordination number of a central metal atom in a complex is determined by
 1) The number of ligands around a metal ion bonded by sigma bonds
 2) the number of only anionic ligands bonded to the metal ion
 3) the number of ligands around a metal ion bonded by sigma and pi-bonds both
 4) the number of ligands around a metal ion bonded by pi-bonds
10. Example showing ionisation isomerism
 1) $[\text{Co}(\text{NO}_3)(\text{NH}_3)_5]\text{SO}_4$ & $[\text{Co}(\text{SO}_4)(\text{NH}_3)_5]\text{NO}_3$
 2) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ & $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 3) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ & $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
 4) $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$ & $[\text{Pt}(\text{NH}_3)_3\text{Cl}][\text{PtCl}_4]$
11. Ligands with which linkage isomerism is possible
 A) NO_2^- B) CN^- C) SCN^-
 1) A only 2) A & B 3) B & C 4) A, B & C
12. $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ are a pair of ----- isomers
 1) Ionization 2) Ligand 3) Co-ordination 4) Hydrate
13. What is wrong about the following pair of compounds? $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ and $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
 1) They are hydration isomers 2) They have different colours
 3) Their 0.1M aqueous solutions have same molar conductivity
 4) They have different IUPAC name
14. Which of the following compounds would exhibit coordination isomerism :
 1) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_2$ 2) $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$ 3) $[\text{Cr}(\text{en})_2]\text{NO}_2$ 4) $[\text{Cr}(\text{H}_2\text{O})_5]\text{Cl}_3$
15. For the empirical formula $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$, the no. of possible coordination isomers would be (ON of pt is +2 in all all isomeric forms)
 1) 1 2) 2 3) 3 4) 4
16. The number of linkage isomers for the compound $\text{K}_2[\text{Cu}(\text{CN})_2(\text{NCS})_2]$ are
 1) 4 2) 6 3) 16 4) 9
17. When two ligands of the same type occupy opposite positions to each other in a coordination polyhedron, the isomer is called
 1) trans- 2) cis- 3) fac- 4) mer-
18. Both geometrical and optical isomerism was shown by
 1) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ 2) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]$ 3) $\text{Pt}[(\text{en})_2\text{Cl}_2]$ 4) $[\text{Pt}(\text{en})_3]$

19. (A) : A cis-isomer has a net dipole moment zero.
 (R) : A cis- isomers has two ligands of the same type occupying adjacent positions.
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false
 4) (A) is false but (R) is true
20. Which of the following compound shows optical isomerism?
 1) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ 2) $[\text{Zn}(\text{Cl})_4]^{2-}$ 3) $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{2-}$ 4) $[\text{Co}(\text{CN})_6]^{3-}$
21. The tris (Ethylene diamine) Cobalt (III) cation $[\text{Co}(\text{en})_3]^{3+}$ can have
 1) Three stereo isomers, all chiral and optically active
 2) Two chiral stereo isomers (enantiomers)
 3) Three stereo isomers all chiral 4) Two stereo isomers both achiral
22. The complex, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ is known to exist in two different coloured forms. This is due to
 1) ionisation isomerism 2) optical isomerism
 3) geometrical isomerism 4) linkage isomerism
23. The d-orbitals involved in sp^3d^2 or d^2sp^3 hybridisation of the central metal ion are :
 1) $d_{x^2-y^2}$ & d_{z^2} 2) d_{xy} & d_{z^2} 3) d_{yz} & $d_{x^2-y^2}$ 4) d_{z^2} only
24. Nickel combines with a uninegative monodentate ligand X to form a paramagnetic complex $[\text{NiX}_4]^{2-}$. The number of unpaired electron/s in the nickel and geometry of this complex ion are respectively
 1) two, square planar 2) one, tetrahedral 3) two, tetrahedral 4) one, square planar
25. What is magnetic moment of $[\text{FeF}_6]^{3-}$?
 1) 6.92 BM 2) 5.92 BM 3) 7.62 BM 4) 3.14 BM
26. Which of the following statements in most likely to be incorrect
 1) $\text{Ti}(\text{NO}_3)_4$ is a colourless compound 2) $[\text{Cu}(\text{NC}_2\text{H}_3)_4]^+\text{BF}_4^-$ is a colourless compound
 3) $[\text{Cr}(\text{NH}_3)_6]^{3+} 3\text{Cl}^-$ is a coloured compound 4) $\text{K}_3[\text{VF}_6]$ is a colourless compound
27. Considering H_2O as a weak field ligand, the no. of unpaired e^- in $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ will be (At.No.-25)
 1) 3 2) 5 3) 2 4) 4
28. Among the following, the species with tetrahedral geometry ?
 1) $[\text{Pd}(\text{CN})_4]^{2-}$ 2) $[\text{Ni}(\text{CN})_4]^{2-}$ 3) $[\text{PdCl}_4]^{2-}$ 4) $[\text{NiCl}_4]^{2-}$
29. The tetrahedral crystal field splitting is only of the octahedral splitting.
 1) 1/2 2) 2/9 3) 4/9 4) 5/9
30. In which of the following complex, the metal ion will have t_{2g}^0, e_g^0 , configuration according to CFT
 1) $[\text{FeF}_6]^{3-}$ 2) $\text{Fe}(\text{CN})_6]^{3-}$ 3) $[\text{Fe}(\text{CN})_6]^{2-}$ 4) none
31. Which of the following is low spin to strong field ligands ?
 1) $d_{xy}^2, d_{xz}^2, d_{yz}^1$ 2) $d_{xy}^2, d_{yz}^1, d_{x^2-y^2}^1, d_{z^2}^1$ 3) $d_{xy}^2, d_{xz}^1, d_{yz}^1, d_{x^2-y^2}^1$ 4) $d_{xy}^1, d_{xz}^1, d_{yz}^1, d_{x^2-y^2}^2$
32. Amongst the following ions which has the highest paramagnetism ?
 1) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ 2) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ 3) $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ 4) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$

Metal carbonyls, Stability and Applications

33. Complex/complex ion 'x' is the most stable amongst following. Then 'x' must be
 1) $\text{Fe}(\text{CO})_5$ 2) $[\text{Fe}(\text{CN})_6]^{-3}$ 3) $[\text{Fe}(\text{C}_2\text{O}_4)_4]^{-3}$ 4) $[\text{Fe}(\text{H}_2\text{O}_6)]^{+3}$
34. From the stability constant (hypothetical value given below predict which is the strongest ligand.
 1) $\text{Cu}^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+}, K = 4.5 \times 10^{11}$
 2) $\text{Cu}^{2+} + 4\text{CN}^- \rightleftharpoons [\text{Cu}(\text{CN})_4]^{2-}, K = 2.0 \times 10^{27}$
 3) $\text{Cu}^{2+} + 2\text{en} \rightleftharpoons [\text{Cu}(\text{en})_2]^{2+}, K = 3 \times 10^{15}$
 4) $\text{Cu}^{2+} + 4\text{H}_2\text{O} \rightleftharpoons [\text{Cu}(\text{H}_2\text{O})_4]^{2+}, K = 9.5 \times 10^8$
35. Which of the following statement is incorrect ?
 1) The stability constant of $[\text{Co}(\text{NH}_3)_6]^{3+}$ is greater than that of $[\text{Co}(\text{NH}_3)_6]^{2+}$
 2) The cyano complexes are more stable than those formed by halide ions
 3) The stability of halide complexes follows the order $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^-$
 4) The stability constant of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is greater than that of $[\text{Cu}(\text{en})_2]^{2+}$
36. M-L bonds in carbonyl compounds posses
 1) only σ character 2) only π character 3) both σ and π character 4) none of the above
37. 0.1M solution of which of the following compounds will have the lowest freezing point ?
 1) $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ 2) $[\text{Co}(\text{Cl})(\text{NH}_3)_5]\text{Cl}_2$
 3) $\text{K}_3[\text{Fe}(\text{CN})_6]$ 4) $[\text{CrCl}_2(\text{NH}_3)]\text{Cl}$

Miscellaneous

38. Which of the following is a tridentate ligand ?
 1) en 2) pn 3) Dien 4) trien
39. $\text{CaCl}_2 \cdot 6\text{NH}_3$ and $\text{CoCl}_2 \cdot 5\text{NH}_3$ do not differ in their
 1) Colour 2) Magnetic property
 3) Number of ionisable chlorides 4) Electrical Conductivity
40. Pick up the incorrect match
 1) NO_2^- : Ambidentate ligand 2) $[\text{NiCl}_2(\text{H}_2\text{O})_4]$: Lewis base is H_2O and Cl^-
 3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$: homoleptic complex 4) $\text{K}_4[\text{Fe}(\text{CN})_6]$: test for K^+ ion
41. Coordination number of the central atom or ion is determined by
 1) Number of sigma bonds 2) Number of pi bonds
 3) Both number of sigma and pi bonds 4) Number of ligands
42. Consider the following ionisable complexes with coordination number 6
 I) $\text{CoCl}_2 \cdot 6\text{NH}_3$ II) $\text{PtCl}_4 \cdot 5\text{NH}_3$
 S_1 Both give white ppt with aq AgNO_3
 S_2 Both have different oxidation number
 S_3 Both are homoleptic complexes
 1) TTF 2) TFT 3) FTT 4) TTT

43. $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ can be distinguished by
 1) Conductance measurement
 2) Using BaCl_2
 3) Using AgNO_3
 4) all of these
44. Identify the incorrect statement about $\text{Fe}(\text{CN})_6^{2-}$
 1) Lewis acid is Fe^{+2}
 2) Lewis base is CN^-
 3) CN^- is ambidentate ligand
 4) Oxidation state of iron is 0
45. Consider the following complex: $[\text{Co}(\text{NH}_3)_5\text{CO}_3]\text{ClO}_4$. The coordination number, oxidation number, number of ions produced in aqueous solution are respectively:
 1) 2,3,6
 2) 2,6,3
 3) 3,2,6
 4) 6,3,2
46. Consider the following complex: $[\text{Ni}(\text{CO})_4]$ let
 P = Atomic number of central metal ion
 Q = Number of Ambidentate ligands
 R = Number of unsymmetrical bidentate ligands
 Then $P + Q + R =$ _____
 1) 1
 2) 0
 3) 2
 4) 4
47. Consider the following ligands Carbonate, Ethylene diamine, Acetyl acetonate, Ethylenediamine-tetra acetic acid, Oxalate.
 P = Number of neutral ligands
 Q = Number of Ambidentate ligands
 R = Number of unsymmetrical bidentate ligands
 Then $P + Q + R =$ _____
 1) 1
 2) 0
 3) 2
 4) 4
48. Oxidation state of Fe in a) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ and
 b) $\text{Na}_2[\text{Fe}(\text{CN})_2\text{NO}]$ is _____ respectively
 1) +1,+2
 2) +2,+1
 3) +2,+2
 4) +2,+3
49. The two compounds $[\text{RuBr}(\text{NH}_3)_3]\text{Cl}$ and $[\text{RuCl}(\text{NH}_3)_3]\text{Br}$ are _____ isomers
 1) Linkage
 2) Ionization
 3) Co-ordination
 4) Geometrical
50. For which of square planar complexes, geometrical isomers are not possible
 1) $[\text{Pt}(\text{NH}_3)_2(\text{C}_2\text{O}_4)]$
 2) $[\text{PdBrCl}(\text{PEt}_3)_2]$
 3) $[\text{IrH}(\text{CO})(\text{PR}_3)_2]$
 4) $[\text{Pd}(\text{gly})_2]$
51. Identify the complex, in which central metal atom is in relatively at high oxidation state
 1) $[\text{Ru}(\text{NH}_3)_5(\text{H}_2\text{O})]\text{Cl}_2$
 2) $[\text{Cr}(\text{NH}_3)_6](\text{NO}_3)_3$
 3) $[\text{Fe}(\text{CO})_5]$
 4) $\text{K}_4[\text{Fe}(\text{CN})_6]$
52. The Cr^{3+} ion forms octahedral complexes with two neutral ligands X and Y. The colour of $[\text{CrX}_6]^{3+}$ is blue while that of $[\text{CrY}_6]^{3+}$ is yellow. Then correct about ligand field strength of X and Y
 1) X is strong
 2) Y is strong
 3) Some of X and Y
 4) Colour is independent of strength of ligand
53. Identify the correct description about $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$
 1) $[\text{NiCl}_4]^{2-}$ is tetrahedral complex
 2) $[\text{NiCl}_4]^{2-}$ is square-planar complex
 3) $[\text{Ni}(\text{CN})_4]^{2-}$ is diamagnetic
 4) $[\text{NiCl}_4]^{2-}$ is low spin complex

54. Which of the following complexes formed by Cu^{2+} ions is most stable
- 1) $\text{Cu}^{2+} + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}$, $\log K = 11.6$
 - 2) $\text{Cu}^{2+} + 4\text{CN}^- \rightarrow [\text{Cu}(\text{CN})_4]^{2-}$, $\log K = 27.3$
 - 3) $\text{Cu}^{2+} + 2(\text{en}) \rightarrow [\text{Cu}(\text{en})_2]^{2+}$, $\log K = 15.4$
 - 4) $\text{Cu}^{2+} + 4\text{H}_2\text{O} \rightarrow [\text{Cu}(\text{H}_2\text{O})_4]^{2+}$, $\log K = 8.9$
55. When 1 mol $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ is treated with excess of AgNO_3 , 3 mol of AgCl are obtained. The formula of the complex is :
- 1) $[\text{CrCl}_3(\text{H}_2\text{O})_3] \cdot 3\text{H}_2\text{O}$
 - 2) $[\text{CrCl}_2(\text{H}_2\text{O})_4] \text{Cl} \cdot 2\text{H}_2\text{O}$
 - 3) $[\text{CrCl}(\text{H}_2\text{O})_5] \text{Cl}_2 \cdot \text{H}_2\text{O}$
 - 4) $[\text{Cr}(\text{H}_2\text{O})_6] \text{Cl}_3$
56. Identify the incorrect statements for the behaviour of ethane-1, 2-diamine as a ligand.
- 1) It is a neutral symmetrical ligand.
 - 2) It is a didentate ligand
 - 3) It is a chelating ligand.
 - 4) It is a weakfield ligand
57. Identify the optically active compounds from the following:
- 1) $[\text{Co}(\text{en})_3]^{3+}$
 - 2) $\text{trans} - [\text{Co}(\text{en})_2 \text{Cl}_2]^+$
 - 3) $[\text{Cr}(\text{NH}_3)_5 \text{SCN}]^{2+}$
 - 4) $[\text{Cr}(\text{NH}_3)_5 \text{Cl}]^{2+}$
58. Consider the violet-colored compound, $[\text{Cr}(\text{OH}_2)_6] \text{Cl}_3$ and the yellow compound, $[\text{Cr}(\text{NH}_3)_6] \text{Cl}_3$. Which of the following statements is false?
- 1) Both chromium metal ions are paramagnetic with 3 unpaired electrons
 - 2) Δ_{oct} for $[\text{Cr}(\text{NH}_3)_6]^{3+}$ is calculated directly from the energy of yellow light.
 - 3) Δ_{oct} for $[\text{Cr}(\text{NH}_2)_6]^{3+}$ is less than Δ_{oct} for $[\text{Cr}(\text{NH}_3)_6]^{3+}$
 - 4) The two complexes absorb their complementary colors.
59. The coordination complex, $[\text{Cu}(\text{OH}_2)_6]^{2+}$ has one unpaired electron. Which of the following statements are true?
- A) The complex is octahedral
 - B) The complex is an outer orbital complex
 - C) The complex is d^2sp^3 hybridized
 - D) The complex is diamagnetic
 - E) The coordination number is 6.
- 1) A,D
 - 2) A,B,E
 - 3) B,C,E
 - 4) B,C
60. In which one of the following species does the transition metal ion have d^3 electronic configuration
- 1) $[\text{Cr}(\text{NH}_3)_6]^{3+}$
 - 2) $[\text{Co}(\text{OH}_2)_6]^{2+}$
 - 3) $[\text{CoF}_6]^{3-}$
 - 4) $[\text{Fe}(\text{CN})_6]^{3-}$
61. When the valence d orbitals of the central metal ion are split in energy in an octahedral ligand field, which orbitals are raised least in energy
- 1) d_{xy} and $d_{x^2-y^2}$
 - 2) d_{xy} , d_{xz} and d_{yz}
 - 3) d_{xz} and d_{yz}
 - 4) d_{xz} , d_{yz} and $d_{x^2-y^2}$
62. For which pair of complexes is the order of values of Δ_{oct} correct
- 1) $[\text{Rh}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{NH}_3)_6]^{3+}$
 - 2) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{Fe}(\text{CN})_6]^{3-}$
 - 3) $[\text{Cr}(\text{OH}_2)_6]^{3+} > [\text{Cr}(\text{OH}_2)_6]^{3+}$
 - 4) $[\text{CrF}_6]^{3-} > [\text{Cr}(\text{CN})_6]^{3-}$
63. The CFSE for a high-spin d^4 octahedral complex is:
- 1) $-0.6\Delta_{\text{oct}}$
 - 2) $-1.8\Delta_{\text{oct}}$
 - 3) $-1.6\Delta_{\text{oct}} + P$
 - 4) $-1.2\Delta_{\text{oct}}$
64. Which of the following correctly places the ligands in their order in the spectrochemical series?
- 1) $\text{Br}^- < \text{Cl}^- < \text{NH}_3 < \text{H}_2\text{O}$
 - 2) $\text{I}^- < \text{Br}^- < \text{H}_2\text{O} < [\text{OH}]^-$
 - 3) $\text{F}^- < \text{Cl}^- < \text{H}_2\text{O} < \text{NH}_3$
 - 4) $\text{I}^- < \text{Cl}^- < \text{H}_2\text{O} < \text{en}$

65. A chelating agent has two or more than two donor atoms to bind to a single metal ion. Which of the following is not a chelating agent?
 1) thiosulphato 2) oxalato 3) glycinate 4) ethane -1,2-diamine
66. The CFSE for octahedral $[\text{CoCl}_6]^{4-}$ is $18,000 \text{ cm}^{-1}$. The CFSE for tetrahedral $[\text{CoCl}_4]^{2-}$ will be
 1) $18,000 \text{ cm}^{-1}$ 2) $16,000 \text{ cm}^{-1}$ 3) $8,000 \text{ cm}^{-1}$ 4) $20,000 \text{ cm}^{-1}$
67. Using valence bond theory, identify the complex ion which is different from other with respect to hybridization of central metal ion
 1) $[\text{Mn}(\text{CN})_6]^{4-}$ 2) $[\text{Co}(\text{NH}_3)_6]^{3+}$ 3) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ 4) $[\text{FeCl}_6]^{4-}$
68. Which is high spin complex?
 1) CoF_6^{3-} 2) $[\text{Co}(\text{NH}_3)_6]^{2+}$ 3) $\text{Fe}(\text{CN})_6^{4-}$ 4) All are correct
69. The EAN of Cr in $[\text{Cr}(\text{SCN})_6]^{3-}$ is :
 1) 35 2) 33 3) 34 4) 37
70. Which order is correct in spectrochemical series for ligands
 1) $\text{Cl}^- < \text{F}^- < \text{C}_2\text{O}_4^{2-} < \text{NO}_2^- < \text{CN}^-$ 2) $\text{CN}^- < \text{C}_2\text{O}_4^{2-} < \text{Cl}^- < \text{NO}_2^- < \text{F}^-$
 3) $\text{C}_2\text{O}_4^{2-} < \text{F}^- < \text{Cl}^- < \text{NO}_2^- < \text{CN}^-$ 4) $\text{F}^- < \text{Cl}^- < \text{NO}_2^- < \text{CN}^- < \text{C}_2\text{O}_4^{2-}$
71. The overall formation constant of the $[\text{Co}(\text{NH}_3)_6]^{2+}$ ion in aqueous solution is 10^5 and the standard potentials for the reduction of $\text{Co}^{3+}(\text{aq})$ and $[\text{Co}(\text{NH}_3)_6]^{3+}(\text{aq})$ are as follows
 $\text{Co}^{3+}(\text{aq}) + \text{e}^- \rightleftharpoons \text{Co}^{2+}(\text{aq}) E^0 = +1.9 \text{ V}$; $[\text{Co}(\text{NH}_3)_6]^{3+}(\text{aq}) + \text{e}^- \rightleftharpoons [\text{Co}(\text{NH}_3)_6]^{2+}(\text{aq}) E^0 = +0.1 \text{ V}$
 Calculate the nearest overall formation constant of the $[\text{Co}(\text{NH}_3)_6]^{3+}$ ion
 1) 10^{30} 2) 10^{33} 3) 10^{35} 4) 10^{38}
72. A compound with empirical formula $\text{Fe}(\text{H}_2\text{O})_4(\text{CN})_2$ has a magnetic moment corresponding to $2\frac{2}{3}$ unpaired electrons per iron. The molecular formula of the complex may be
 1) $[\text{Fe}(\text{H}_2\text{O})_4(\text{CN})_2]$ 2) $[\text{Fe}(\text{H}_2\text{O})_6][\text{Fe}(\text{CN})_6]$
 3) $[\text{Fe}(\text{H}_2\text{O})_6]_4[\text{Fe}(\text{CN})_6]_3$ 4) $[\text{Fe}(\text{H}_2\text{O})_6]_2[\text{Fe}(\text{CN})_6]$
73. Arrange in increasing order of acidic nature
 I) $[\text{Na}(\text{H}_2\text{O})_6]^+$ II) $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ III) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ IV) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
 1) IV < III < II < I 2) I < II < III < IV 3) I < III < IV < II 4) IV < II < III < I

Numerical Value Type Questions

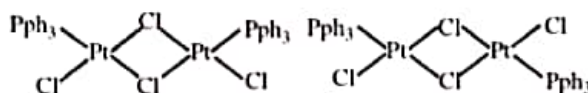
74. The number of chloride ions which would be precipitated, when $\text{CrCl}_3 \cdot 4\text{NH}_3$ is treated with silver nitrate solution
75. The primary valency of the central transition metal ion in a complex compound $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
76. The number of geometrical isomers of $[\text{Co}(\text{NH}_3)_3(\text{NO}_3)_3]$ is
77. The number of isomers possible for square planar complex $\text{K}_2[\text{PdClBr}_2\text{SCN}]$ are
78. Number of isomers possible for $[\text{Co}(\text{en})_2\text{Cl}(\text{NO}_2)]\text{Cl}$ are
79. The possible no. of coordination isomers of $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$ is

EXERCISE-II

Straight Objective Type Questions

IUPAC, Wernes theory, isomerism

- IUPAC name of $\text{Ni}(\text{CO})_4$ is
 - tetracarbonylNickel(II)
 - tetracarbonylNickel (O)
 - tetracarbonylNickelate (II)
 - tetracarbonylNickelate (O)
- Effective atomic number of Cobalt in the complex ion $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ is
 - 27
 - 36
 - 33
 - 35
- Stable complex based on EAN rule
 - $\text{K}_4[\text{Fe}(\text{CN})_6]$
 - $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 - $[\text{Ni}(\text{CO})_4]$
 - $\text{K}_2[\text{Ni}(\text{CN})_4]$
 - i only
 - i & ii only
 - i, ii & iii only
 - all
- Which of the following isomerism, exhibited by octahedral $[\text{Co}(\text{NH}_3)_4\text{Br}_2]\text{Cl}$?
 - Geometrical and ionization
 - Geometrical and optical
 - Optical and ionization
 - Geometrical only
- Fac and Mer isomerism is associated with which of the following general formula?
 - $[\text{M}(\text{AA})_2]$
 - $[\text{M}(\text{AA})_3]$
 - $[\text{MABCD}]$
 - $[\text{MA}_3\text{B}_3]$
- Complexes given below show :

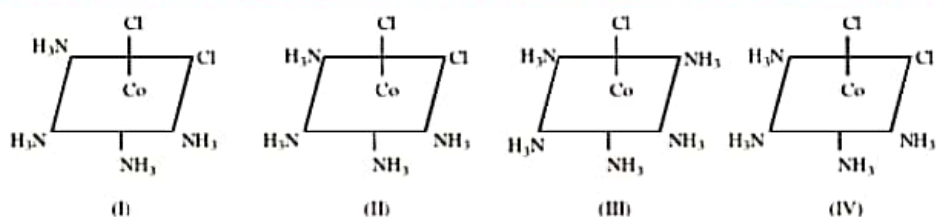


- Optical isomerism
 - Coordinate isomerism
 - Geometrical isomerism
 - Bridge isomerism
- The optically active species among the following is :
 - $[\text{Cr}(\text{NH}_3)_6]^{3+}$
 - $[\text{Co}(\text{CN})_6]^{3-}$
 - $[\text{Co}(\text{gly})_3]$
 - $[\text{Ru}(\text{NH}_3)_6]^{3+}$
 - Which of the following complex ion has a pair of enantiomers ?
 - $[\text{Zn}(\text{acac})(\text{en})_2]^+$
 - $[\text{PtBrCl}(\text{CN})(\text{SCN})]^{2-}$
 - $[\text{Ir}(\text{acac})_2(\text{H}_2\text{O})(\text{NH}_3)]^+$
 - $[\text{CrCl}_3(\text{SCN})(\text{H}_2\text{O})]^-$
 - Select incorrect statement about complex $[\text{Cr}(\text{NH}_3)_5(\text{NO}_2)][\text{Zn}(\text{SCN})_4]$.
 - It show coordination isomerism
 - It show optical isomerism
 - It show linkage isomerism
 - IUPAC name is pentaammine nitrito-N-Chromium (III) tetrathiocyanato-S-zincate(II)

IUPAC, Wernes theory, Isomerism

- The primary valency of 'Fe' in the complex $\text{K}_4[\text{Fe}(\text{CN})_6]$ is
 - 2
 - 3
 - 6
 - 4
- One mole of the complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of $\text{AgCl}(\text{s})$. The structure of the complex is
 - $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 - $[\text{Co}(\text{NH}_3)_3\text{Cl}_3].2\text{NH}_3$
 - $[\text{Co}(\text{NH}_3)_4\text{Cl}_2].\text{Cl}.\text{NH}_3.4$
 - $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
- The secondary valency of Chromium in $[\text{Cr}(\text{en})_3]\text{Cl}_3$ is
 - 6
 - 3
 - 2
 - 4

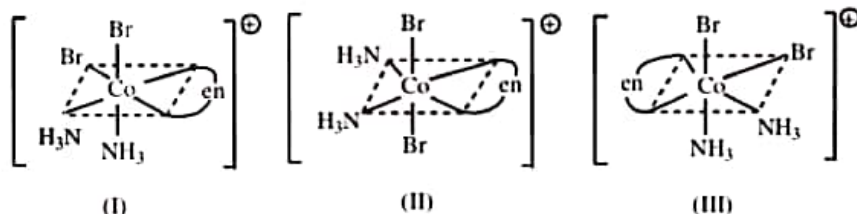
13. When 1 mole of $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ is added to excess of AgNO_3 solution the weight of AgCl precipitated is
 1) 143.5g 2) 108 g 3) zero 4) 54 g
14. A complex of Co^{+3} having molecular formula $\text{CoCl}_3 \cdot x\text{NH}_3$ when dissolved in water three ions are produced. The number of Cl^- ions satisfying both primary and secondary valencies is
 1) 3 2) 1 3) 4 4) zero
15. Which one of the following pairs of isomers and types of isomerism are correctly matched?
 (i) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{Cl}_2$ (Linkage)
 (ii) $[\text{Cu}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_4][\text{CuCl}_4]$ (Coordination)
 (iii) $[\text{PtCl}_2(\text{NH}_3)_4]\text{Br}_2$ and $[\text{PtBr}_2(\text{NH}_3)_4]\text{Cl}_2$ (Ionization)
 1) (ii) and (iii) 2) (i), (ii) and (iii) 3) (i) and (iii) 4) (i) and (ii)
16. Theoretically, the number of geometrical isomers expected for octahedral complex, $[\text{Mabedef}]$ is
 1) 0 2) 15 3) 12 4) 30
17. Both geometrical and optical isomerisms are shown by
 1) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ 2) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ 3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ 4) $[\text{Cr}(\text{Ox}_3)]^{3-}$
18. Which one of the following exhibit chirality ?
 1) $[\text{Cr}(\text{Ox}_3)]^{3-}$ 2) $\text{Cis}-[\text{PtCl}_2(\text{en})]$ 3) $\text{Cis}-[\text{RhCl}_2(\text{NH}_3)_2]^{-1}$ 4) $\text{Mer}-[\text{Co}(\text{NO}_2)_3(\text{trien})]$
19. Consider the following arrangements of the octahedral complex ion $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$



which of the following statements is incorrect regarding these structure.

- 1) I, II are enantiomers 2) II, III are trans and Cis isomers respectively
 3) III, IV are trans and Cis isomers respectively 4) II, IV have identical structure
20. In the complex $\text{MCl}_3 \cdot 5\text{H}_2\text{O}$, the Co-ordination number of the metal M is six and there is no molecule of hydration. Then the volume of 0.1M AgNO_3 solution needed to precipitate the free chloride ions in 200ml of 0.01M solution of the complex is
 1) 80 ml 2) 40 ml 3) 20 ml 4) 120 ml
21. How many geometrical isomers are possible for complex $[\text{Mab}(\text{AB})_2]^{n\pm}$?
 1) 5 2) 4 3) 3 4) 6
22. Which of the following isomerism, exhibited by $[\text{CrCl}_2(\text{OH})_2(\text{NH}_3)_2]^{-7}$
 1) Ionization 2) Geometrical 3) Hydrate 4) Linkage
23. Which of the following will have two stereoisomeric forms?
 I) $[\text{Cr}(\text{NO}_3)_3(\text{NH}_3)_3]$ II) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ III) $[\text{CoCl}_2(\text{en})_2]^+$ IV) $[\text{CoBrCl}(\text{ox})_2]^{3-}$
 1) I only 2) I and II 3) III and IV 4) All of these

24. Three arrangements are shown for the complex $[\text{CoBr}_2(\text{NH}_3)_2(\text{en})]^\oplus$. Which one is wrong statement?



- 1) I and II are geometrical isomers
2) II and III are optically active isomers
3) I and III are optically active isomers
4) II and III are geometrical isomers
25. A metal complex of coordination number six having three different types of ligands a, b and c of composition $\text{Ma}_2\text{b}_2\text{c}_2$ can exist in several geometrical isomeric forms; the total number of such isomers is :

- 1) 3 2) 5 3) 7 4) 9

26. The complex with a maximum number of stereoisomers is :

- 1) $[\text{PtCl}_3(\text{C}_2\text{H}_4)]^-$ 2) $[\text{CuBr}_2\text{Cl}_2]^{2-}$ 3) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ 4) $[\text{Cr}(\text{NH}_3)_2(\text{en})_2]^{3+}$

27. Which of the following complex is non-ionisable and has two geometrical isomers ?

- 1) $\text{PtCl}_4 \cdot 3\text{NH}_3$ 2) $\text{PtCl}_4 \cdot 6\text{NH}_3$ 3) $\text{PtCl}_4 \cdot 2\text{NH}_3$ 4) $\text{PtCl}_4 \cdot 4\text{NH}_3$

VBT and CFT

28. There are four complexes of Ni. Select the complex(es) which will be attracted by magnetic field :

- (I) $[\text{Ni}(\text{CN})_4]^{2-}$ (II) $[\text{NiCl}_4]^{2-}$ (III) $\text{Ni}(\text{CO})_4$ (IV) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
1) I only 2) IV only 3) II, III and IV 4) II and IV

29. Which of the following complex is an outer orbital complex ?

- 1) $[\text{Ni}(\text{NH}_3)_6]^{2+}$ 2) $[\text{Mn}(\text{CN})_6]^{4-}$ 3) $[\text{Co}(\text{NH}_3)_6]^{3+}$ 4) $[\text{Fe}(\text{CN})_6]^{4-}$

30. Which of the following order is correct in spectrochemical series of ligands ?

- 1) $\text{Cl}^- < \text{F}^- < \text{C}_2\text{O}_4^{2-} < \text{NO}_2^- < \text{CN}^-$ 2) $\text{NO}_2^- < \text{C}_2\text{O}_4^{2-} < \text{Cl}^- < \text{F}^- < \text{CN}^-$
3) $\text{C}_2\text{O}_4^{2-} < \text{F}^- < \text{Cl}^- < \text{NO}_2^- < \text{CN}^-$ 4) $\text{F}^- < \text{Cl}^- < \text{NO}_2^- < \text{CN}^- < \text{C}_2\text{O}_4^{2-}$

31. The species with spin only magnetic moment of $\sqrt{24}$ BM is :

- 1) $[\text{CoF}_3(\text{H}_2\text{O})_3]$ 2) $[\text{CoCl}_4]^{2-}$ 3) $[\text{NiCl}_4]^{2-}$ 4) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$

32. Which of the following is correctly matched?

- | Column - I | Column - II | Column - III |
|-------------------------------|--------------|------------------------------------|
| 1) $[\text{Cr}(\text{CO})_6]$ | paramagnetic | Octahedral sp^3d^2 |
| 2) $[\text{Fe}(\text{CO})_5]$ | paramagnetic | Trigonal bipyramid sp^3 |
| 3) $[\text{Co}(\text{CO})_4]$ | Diamagnetic | Tetrahedral, sp^3 |
| 4) $[\text{Ni}(\text{CO})_4]$ | Diamagnetic | Square plane, dsp^3 |

33. Which of the following statement is not true for the reaction given below?



- 1) It is a ligand substitution reaction
 - 2) NH_3 is a relatively strong field ligand while H_2O is a weak field ligand
 - 3) During the reaction, there is a change in colour from light blue to dark blue
 - 4) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ has a tetrahedral structure and is paramagnetic
34. The correct order of magnetic moments (spin values in BM) among is :
- 1) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
 - 2) $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
 - 3) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$
 - 4) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$
35. What is electronic arrangement of metal atom/ion in octahedral complex with d^4 configuration, if $\Delta_0 < \text{pairing energy}$?
- 1) $t_{2g}^4 e_g^0$
 - 2) $e_g^4 t_{2g}^0$
 - 3) $t_{2g}^3 e_g^1$
 - 4) $e_g^2 t_{2g}^2$
36. All the following complex ions are found to be paramagnetic
 $\text{P} = [\text{FeF}_6]^{3-}$ $\text{Q} = [\text{CoF}_6]^{3-}$ $\text{R} = [\text{V}(\text{H}_2\text{O})_6]^{3+}$ $\text{S} = [\text{T}(\text{H}_2\text{O})_6]^{3+}$
 The correct order of their paramagnetic moment (spin only) is
- 1) $\text{P} > \text{Q} > \text{R} > \text{S}$
 - 2) $\text{P} < \text{Q} < \text{R} < \text{S}$
 - 3) $\text{P} = \text{Q} = \text{R} = \text{S}$
 - 4) $\text{P} > \text{R} > \text{Q} > \text{S}$
37. For which of the following types of ions the no. of unpaired e^- in octahedral complexes fixed at the same no. as in the free ion no matter how weak or strong the crystal field is ?
- 1) d^3
 - 2) d^4
 - 3) d^5
 - 4) d^6
38. Amongst I = $[\text{Co}(\text{Ox})_3]^{3-}$, II = $[\text{CoF}_6]^{3-}$, III = $[\text{Co}(\text{NH}_3)_6]^{3+}$
- 1) I, II are para, and III is dia
 - 2) I, III are para, and II is dia
 - 3) I, III are dia, and II is para
 - 4) II, III are para, and I is dia
39. All of the following complex show decrease in their weight when placed in magnetic balance then the group of complexes having tetrahedral geometry is
- I) $\text{Ni}(\text{CO})_4$
 - II) $\text{K}[\text{AgF}_4]$
 - III) $\text{Na}_2[\text{Zn}(\text{CN})_4]$
 - IV) $\text{K}_2[\text{Pt}(\text{Cl})_4]$
 - V) $[\text{RhCl}(\text{pPh}_3)_3]$
- 1) II, III, IV
 - 2) I, II, III
 - 3) I, III, IV
 - 4) None
40. On treatment of $[\text{Ni}(\text{NH}_3)_4]^{+2}$ with conc. HCl two compounds I and II having the same formula $[\text{Ni}(\text{NH}_3)_2\text{Cl}_2]$ are obtained. I can be converted into II by boiling with dil. HCl . A solution of I reacts with oxalic acid to form $[\text{Ni}(\text{NH}_3)_2(\text{C}_2\text{O}_4)]$ where as II does not react. Point out the correct statement from the following
- 1) I cis, II trans, both tetrahedral
 - 2) I cis, II trans both square planar
 - 3) I trans, II cis, both tetrahedral
 - 4) I trans, II cis both square planar
41. Identify the incorrect statement
- 1) Δ_0 increasing order is $[\text{CrCl}_6]^{3-} < [\text{Cr}(\text{NH}_3)_6]^{3+} < [\text{Cr}(\text{CN})_6]^{3-}$
 - 2) CFSE for $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is $-0.4\Delta_0$
 - 3) $[\text{NiCl}_4]^{2-}$ as well as $[\text{Ni}(\text{CO})_4]$ are paramagnetic
 - 4) The halide ions are arranged as $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^-$ in the spectro chemical series

42. Among the following, the ions having same magnetic moment are
 1) $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ 2) $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ 3) both (a) and (b) 4) None
43. Which of the following is correctly matched?
 1) $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Fe}(\text{CN})_6]^{3-}$ — both are octahedral and diamagnetic with d^2sp^3 hybridisation
 2) $\text{Ni}(\text{CO})_4$ and $[\text{Ni}(\text{CN})_4]^-$ — both are tetrahedral and diamagnetic with sp^3 hybridisation
 3) $\text{Ni}(\text{CO})_4$ and $[\text{Co}(\text{CO})_4]^{3-}$ — both are tetrahedral and diamagnetic
 4) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ and $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ — both are paramagnetic and metal is d^2sp^3 hybridised
44. The hybridization of the complex $[\text{CrCl}_2(\text{NO}_2)_2(\text{NH}_3)_2]^+$ is :
 1) sp^3d^2 2) d^2sp^3 3) sp^3d 4) cannot be predicted
45. The CFSE for $[\text{CoCl}_6]^{4-}$ complex is 18000 cm^{-1} . The Δ for $[\text{CoCl}_4]^{2-}$ will be :
 1) 18000 cm^{-1} 2) 16000 cm^{-1} 3) 8000 cm^{-1} 4) 2000 cm^{-1}
46. Which of the following order of CFSE is incorrect?
 1) $[\text{Co}(\text{en})_3]^{3+} > [\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{H}_2\text{O})_6]^{3+}$ 2) $[\text{PtCl}_4]^{2-} > [\text{PdCl}_4]^{2-} > [\text{NiCl}_4]^{2-}$
 3) $[\text{Ni}(\text{DMG})_2] < [\text{Ni}(\text{en})_2]^{2+}$ 4) $[\text{Co}(\text{ox})_3]^{3-} < [\text{Co}(\text{en})_3]^{3+}$
47. For which of the following d^n configuration of octahedral complexes, can not exist in both high spin and low spin forms :
 I) d^3 II) d^5 III) d^6 IV) d^8
 1) I, II and III 2) II, III and IV 3) I and IV 4) None of these
48. A $[\text{M}(\text{H}_2\text{O})_6]^{2+}$ complex typically absorbs at around 600 nm. It is allowed to react with ammonia to form a new complex $[\text{M}(\text{NH}_3)_6]^{2+}$ that should have absorption at :
 1) 800 nm 2) 580 nm 3) 620 nm 4) 320 nm

Metal carbonyls, Stability and Applications

49. Ferrous sulphate forms Turnbull's blue with
 1) KCN 2) NH_4SCN 3) $\text{K}_3[\text{Fe}(\text{CN})_6]$ 4) $\text{K}_4[\text{Fe}(\text{CN})_6]$
50. The ligands in anticancer drug "cis-platin" are
 1) NH_3 , Cl 2) NH_3 , H_2O 3) Cl, H_2O 4) NO, Cl
51. Which of the following carbonyl is less stable ?
 1) $\text{Mn}_2(\text{CO})_{10}$ 2) $\text{V}(\text{CO})_6$ 3) $\text{Ni}(\text{CO})_4$ 4) $\text{Fe}(\text{CO})_5$
52. From the stability constant (hypothetical values given below) predict which is the strongest ligand
 1) $\text{Cu}^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+}$ $K = 4.5 \times 10^{11}$ 2) $\text{Cu}^{2+} + 4\text{CN}^- \rightleftharpoons [\text{Cu}(\text{CN})_4]^{2-}$ $K = 2.0 \times 10^{27}$
 3) $\text{Cu}^{2+} + 2\text{en} \rightleftharpoons [\text{Cu}(\text{en})_2]^{2+}$ $K = 3 \times 10^{15}$ 4) $\text{Cu}^{2+} + 4\text{H}_2\text{O} \rightleftharpoons [\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ $K = 9.5 \times 10^8$

53. Which of the following statements is incorrect ?

- 1) The stability constant of $[\text{Co}(\text{NH}_3)_6]^{3+}$ is greater than that of $[\text{Co}(\text{NH}_3)_6]^{2+}$
- 2) The cyano complexes are more stable than those formed by halide ions
- 3) The stability of halide complexes follows the order $\text{I}^- < \text{Br}^- < \text{Cl}^- < \text{F}^-$
- 4) The stability constant of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is greater than that of $[\text{Cu}(\text{en})_2]^{2+}$

54. Which of the following is a correct Irving - Williams order ?

- 1) $\text{Mn}^{2+} < \text{Fe}^{2+} < \text{Co}^{2+} < \text{Ni}^{2+}$
- 2) $\text{Ni}^{2+} < \text{Co}^{2+} < \text{Fe}^{2+} < \text{Mn}^{2+}$
- 3) $\text{Fe}^{2+} < \text{Mn}^{2+} < \text{Ni}^{2+} < \text{Co}^{2+}$
- 4) $\text{Co}^{2+} < \text{Mn}^{2+} < \text{Fe}^{2+} < \text{Ni}^{2+}$

55. Which of the following is arranged in order of increasing stability of the metal complexes ? (Oxidation states are given in parenthesis)

- 1) $\text{Ag (I)} < \text{Pt (III)} < \text{Cr (III)} < \text{Co (III)}$
- 2) $\text{Co (III)} < \text{Cr (III)} < \text{Pt (II)} < \text{Ag (I)}$
- 3) $\text{Cr (III)} < \text{Pt (II)} < \text{Ag (I)} < \text{Co (III)}$
- 4) $\text{Pt (II)} < \text{Co (III)} < \text{Ag (I)} < \text{Cr (III)}$

56. Which of the following statements are correct about stability of chelates ?

- 1) As the no. of rings in complex increases, stability of chelate also increases
- 2) A chelate having five memberd ring on more stable if it contains double bonds
- 3) A chelate having six memberd ring in more stable if it does not contain double bonds
- 4) Chelating ligands are atleast monodentate ligands

57. Assertion : Oxidation state of Fe in $\text{Fe}(\text{CO})_5$ is zero.

Reason : Synergic bonding takes place in this metal carbonyl complex.

58. Assertion : Zeise's salt is a π -bonded organometallic compound.

Reason : Zeise's salt contain C_2H_4 molecules as one of the ligand.

59. Which complex is most stable?

- 1) $[\text{Cu}(\text{CN})_2]^-$ $K_d = 1 \times 10^{-16}$
- 2) $[\text{Fe}(\text{CN})_6]^{4-}$ $K_d = 1 \times 10^{-37}$
- 3) $[\text{Fe}(\text{CN})_6]^{2-}$ $K_d = 1 \times 10^{-44}$
- 4) $[\text{Ag}(\text{CN})_2]^-$ $K_d = 1 \times 10^{-20}$

60. In the isoelectronic series of metal carbonyl, the CO bond strength is expected to increase in the order .

- 1) $[\text{Mn}(\text{CO})_6]^+ < [\text{Cr}(\text{CO})_6] < [\text{V}(\text{CO})_6]^-$
- 2) $[\text{V}(\text{CO})_6]^- < [\text{Mn}(\text{CO})_6]^+ < [\text{Cr}(\text{CO})_6]$
- 3) $[\text{V}(\text{CO})_6]^- < [\text{Cr}(\text{CO})_6] < [\text{Mn}(\text{CO})_6]^+$
- 4) $[\text{Cr}(\text{CO})_6] < [\text{Mn}(\text{CO})_6]^+ < [\text{V}(\text{CO})_6]^-$

61. Which is not true about metal carbonyls ?

- 1) Here CO acts as a Lewis base as well as Lewis acid
- 2) Here metal acts as Lewis base as well as Lewis acid
- 3) Here $d\pi - p\pi$ back bonding takes place
- 4) Here $d\pi - p\pi$ back bonding takes place

62. The compound that is not olefinic organometallic is

- 1) $\text{K}[\text{C}_2\text{H}_4\text{PtCl}_3] \cdot 3\text{H}_2\text{O}$
- 2) $\text{Be}(\text{CH}_2)_2$
- 3) $\text{K}[\text{C}_2\text{H}_4\text{PtCl}_3]$
- 4) $\text{C}_4\text{H}_4\text{Fe}(\text{CO})_3$

63. Which of the following is the increasing order of acidity ?

- 1) $[\text{Na}(\text{H}_2\text{O})_6]^+ < [\text{Mn}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Sc}(\text{H}_2\text{O})_6]^{3+}$
- 2) $[\text{Sc}(\text{H}_2\text{O})_6]^{3+} < [\text{Mn}(\text{H}_2\text{O})_6]^{2+} < [\text{Na}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
- 3) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+} < [\text{Na}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Sc}(\text{H}_2\text{O})_6]^{3+}$
- 4) $[\text{Na}(\text{H}_2\text{O})_6]^+ < [\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Mn}(\text{H}_2\text{O})_6]^{2+} < [\text{Sc}(\text{H}_2\text{O})_6]^{3+}$

64. What is the common formula of Turnbull's blue and Prussian blue?

- 1) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$
- 2) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
- 3) $\text{KFe}[\text{Fe}(\text{CN})_6]$
- 4) $\text{KFe}_2[\text{Fe}(\text{CN})_6]$

65. Ti^{3+} , V^{3+} , Fe^{3+} and Co^{2+} can form tetrahedral complexes but Cr^{3+} never form tetrahedral complexes. This is because.

- 1) Crystal field stabilization energy in octahedral vis - a - vis tetrahedral Cr^{3+} system plays the deciding role
- 2) Cr^{3+} forces high crystal field splitting with a variety of ligands .
- 3) Electronegativity of Cr^{3+} is the largest among these trivalent 3d - metals and so chromium prefers to be associated with as many ligands as its radius permits .
- 4) Both (b) and (c)

66. Both Co^{3+} and Pt^{4+} have a coordination number of six. Which of the following pairs of complexes will show approximately the same electrical conductance for their 0.001 M aqueous solutions ?

- 1) $\text{CoCl}_3 \cdot 6\text{NH}_3$ and $\text{PtCl}_4 \cdot 5\text{NH}_3$
- 2) $\text{CoCl}_2 \cdot 6\text{NH}_3$ and $\text{PtCl}_4 \cdot 3\text{NH}_3$
- 3) $\text{CoCl}_3 \cdot 5\text{NH}_3$ and $\text{PtCl}_4 \cdot 6\text{NH}_3$
- 4) $\text{CoCl}_3 \cdot 4\text{NH}_3$ and $\text{PtCl}_4 \cdot 4\text{NH}_3$

67. Which of the following chromium complex is most stable.

- 1) $[\text{Cr}(\text{CN})_6]^{3-}$
- 2) $[\text{Cr}(\text{NH}_3)_6]^{3+}$
- 3) $[\text{Cr}(\text{en})_3]^{3+}$
- 4) $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$

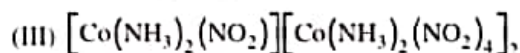
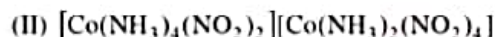
68. Why $[\text{Ni}(\text{en})_3]^{2+}$ is nearly 10^{10} times more stable than $[\text{Ni}(\text{NH}_3)_6]^{2+}$?

- 1) NH_3 is a weak ligand
- 2) 'en' forms more stable complex by increasing entropy
- 3) Six NH_3 ligands cause steric hindrance around the Ni^{2+} centre
- 4) NH_3 evaporates easily and causes instability to the $[\text{Ni}(\text{NH}_3)_6]^{2+}$ complex

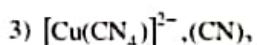
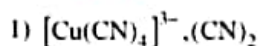
69. Which of the following is not the correct systematic name of corresponding compound

- 1) $\text{Ni}(\text{CO})_2(\text{PPh}_3)_2$: Dicarbonyl bis (triphenylphosphine) nickel (0)
- 2) $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$: Sodium pentacyanonitrososodium ferrate(II)
- 3) $\text{NaMn}(\text{CO})_5$: Sodium pentacarbonyl manganate (I)
- 4) $\text{K}[\text{Pt}(\text{C}_2\text{H}_4)\text{Cl}_3]$: Potassium trichlorido (η^2 -ethylene) platinate (II)

70. Consider the following complexes



- Freezing points of the solutions of each complex containing 24.8 gm solute per kg water $[K_f(\text{H}_2\text{O}) = 1.88^\circ\text{C/m}]$ will be in the order I>II>III
 - Freezing points of the solutions of each complex containing 24.8 gm solute per kg water $[k_f(\text{H}_2\text{O}) = 1.86^\circ\text{C/m}]$ will be in the order I<II<III
 - Oxidation number of Cobalt is +3 in all coordination spheres (I), (II) and (III)
 - Complex (I) can show only two isomers facial and meridional
71. $\text{CuSO}_4(\text{aq}) \xrightarrow{\text{H}_2\text{S}} \text{A} \downarrow \xrightarrow{\text{Excess}} \text{B} + \text{C}$ then final product B and C are respectively



72. Metal M belongs to 4th period and VIII group and forms a complex $\text{K}_3[\text{M}(\text{CN})_6]$ with magnetic moment 1.73 BM, then EAN of $[\text{M}(\text{CO})_5]$ is

1) 35

2) 36

3) 37

4) 54

73. If Δ_0 is the octahedral splitting energy of d orbitals of a metal ion with d^4 configuration, the stabilisation energy in a weak field and in a strong field are respectively

1) $-0.8\Delta_0$ and $-1.8\Delta_0$

2) $-0.6\Delta_0$ and $-1.6\Delta_0$

3) $-0.4\Delta_0$ and $-2.4\Delta_0$

4) $-1.2\Delta_0$ and $-1.2\Delta_0$

Numerical Value Type Questions

74. Total number of space (stereo) isomers of complex ion $[\text{Cr}(\text{gly})(\text{en})_2]^{2+}$ are.....
75. The number of unpaired electrons in d^7 , low spin, octahedral complex is
76. The number of unpaired electron in the complex ion $[\text{CoF}_6]^{3-}$ is
77. Maximum number of dative bonds that can be formed by EDTA
78. No. of unpaired e^- in $\text{K}[\text{Ag}(\text{CN})_2]$ is
79. A complex is prepared by mixing CoCl_2 and NH_3 . 0.1M aq solution of the complex was found to freeze at -0.372°C . How many no. of ions are present in solution by formed complex compound (K_f of water is $1.86 \text{ K.Kg mole}^{-1}$)

KEY SHEET

EXERCISE-I

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1) 3 | 2) 1 | 3) 2 | 4) 2 | 5) 1 | 6) 3 | 7) 1 | 8) 4 | 9) 1 | 10) 1 |
| 11) 4 | 12) 1 | 13) 3 | 14) 2 | 15) 2 | 16) 1 | 17) 1 | 18) 3 | 19) 4 | 20) 3 |
| 21) 2 | 22) 3 | 23) 1 | 24) 3 | 25) 2 | 26) 4 | 27) 2 | 28) 4 | 29) 3 | 30) 3 |

31) 1	32) 2	33) 3	34) 2	35) 4	36) 3	37) 1	38) 3	39) 2	40) 3
41) 1	42) 1	43) 4	44) 4	45) 4	46) 2	47) 1	48) 1	49) 2	50) 1
51) 2	52) 2	53) 3	54) 2	55) 4	56) 4	57) 1	58) 2	59) 2	60) 1
61) 2	62) 1	63) 1	64) 4	65) 1	66) 3	67) 4	68) 1	69) 2	70) 1
71) 3	72) 4	73) 3	74) 1	75) 3	76) 2	77) 4	78) 9	79) 6	

EXERCISE-II

1) 2	2) 2	3) 3	4) 1	5) 4	6) 3	7) 3	8) 3	9) 2	10) 1
11) 1	12) 1	13) 3	14) 2	15) 2	16) 2	17) 1	18) 1	19) 1	20) 2
21) 4	22) 2	23) 2	24) 2	25) 2	26) 4	27) 3	28) 4	29) 2	30) 1
31) 1	32) 2	33) 4	34) 4	35) 3	36) 1	37) 1	38) 3	39) 4	40) 2
41) 3	42) 3	43) 3	44) 2	45) 3	46) 3	47) 3	48) 2	49) 3	50) 1
51) 2	52) 2	53) 4	54) 1	55) 1	56) 1	57) 2	58) 1	59) 3	60) 3
61) 4	62) 4	63) 1	64) 3	65) 1	66) 1	67) 1	68) 2	69) 3	70) 3
71) 1	72) 3	73) 2	74) 2	75) 1	76) 4	77) 6	78) 1	79) 2	

LEVEL-II (ADVANCED)

More than One correct answer Type Questions

- Which of the following statements is not true about the complex ion $[\text{CrCl}(\text{NO}_2)(\text{en})_2]^+$?
 - It has two geometrical isomers cis and trans.
 - cis and trans forms are not diastereomers to each other
 - Only the cis isomer displays optical activity
 - It has three optically active isomers : d, l and meso forms.
- The compound that exhibit geometrical isomerism is
 - $[\text{Pt}(\text{en})\text{Cl}_2]$
 - $[\text{Pt}(\text{en})_2]\text{Cl}_2$
 - $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{Cl}_2$
 - $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
- Which of the following statement is false.
 - In $[\text{PtCl}_2(\text{NH}_3)_4]^{+2}$ complex ion, the cis-form is optically inactive
 - In $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$, geometrical isomerism does not exist, while optical isomerism exists
 - In $[\text{Mabed}]^{n\pm}$ tetrahedral complexes, optical isomerism cannot be observed
 - In $[\text{Mabed}]^{n\pm}$ square planar complexes, optical isomerism can be observed
- Which of the following statement is true about the complex $[\text{CrCl}_3(\text{OH})_2(\text{NH}_3)]^{2-}$ ion ?
 - It has 3 geometrical isomers
 - Only one space isomer is optically active and remaining are inactive
 - There are total four space isomers
 - The magnetic moment of complex ion is 3.89 BM

15. Which is high spin complex :
 a) CoCl_6^{3-} b) FeF_6^{3-} c) $\text{Co}(\text{NH}_3)_6^{3+}$ d) $[\text{Ni}(\text{CN})_4]^{2-}$
16. A d-block element forms octahedral complex but its magnetic moment remains same either in strong field or in weak field ligand. Which of the following is/are correct ?
 a) Element always forms colourless compound
 b) Number of electrons in t_{2g} orbitals are higher than in e_g orbitals
 c) It can have either d^3 or d^8 configuration
 d) It can have either d^7 or d^8 configuration
17. For which of the following d^n configuration of octahedral complex(es), cannot exist in both high spin and low spin forms ?
 a) d^1 b) d^5 c) d^6 d) d^8
18. Which one of the following statement(s) is/are false ?
 a) Weak ligand like F^- , Cl^- and OH^- usually form low spin complexes
 b) Strong ligands like CN^- and NO_2^- , generally form high spin complexes
 c) $[\text{FeF}_6]^{3-}$ is high spin complexes d) $[\text{Ni}(\text{CO})_4]$ is high spin complex
19. The correct statements among the following are
 a) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ differ in the magnetic property
 b) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ differ in the geometry
 c) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ differ in primary valencies of nickel
 d) The complexes $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ differ in the state of hybridisation
20. Which of the following is a chelate ?
 a) Cis-platin - a drug used in the treatment of cancer
 b) Haemoglobin - a protein present in blood
 c) Chlorophyll - a green plant pigment which acts as a photosensitiser in the synthesis of carbohydrates
 d) Vitamin B_{12} - It is a cobalt (III) complex

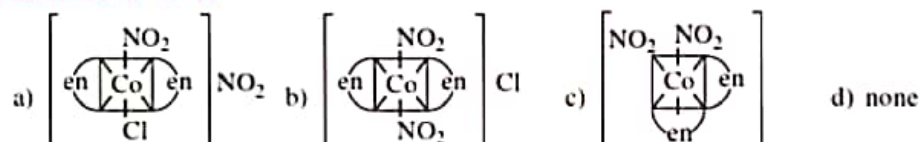
Linked Comprehension Type Questions

Passage - I :

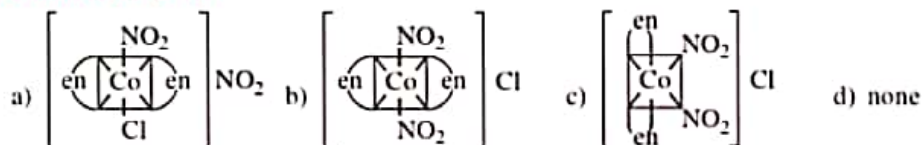
A compound $[\text{Co}(\text{en})_2(\text{NO}_2)_2]\text{Cl}$ exists in different isomeric forms with their characteristic properties as given below

Property	A	B	C
i) optical activity	×	×	✓
ii) reaction with AgNO_3	×	✓	✓
iii) reaction with en	×	×	✓

21. Structure of 'A' is



22. Structure of 'B' is



Passage - II :

Complex compounds can exhibit isomerism like structural and stereo. The complexes containing coordination No, 1, 2, 3 cannot exhibit geometrical isomerism. Geometrical and optical isomerism is common in the complex having coordination number 4 and 6. Ma_3b_3 shows facial and meridional isomerism.

23. Number of geometrical isomers possible for complex $[\text{Pt}(\text{NH}_3)(\text{Py})(\text{Cl})(\text{Br})]$ is

- a) 2 b) 3 c) 1 d) 4

24. $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$ are

- a) linkage isomers b) ionisation isomers
c) coordinate isomers d) geometrical isomers

Passage-III :

The dissociation of complex may be expressed as $[\text{ML}_x]^{4+} \rightleftharpoons \text{M}^{4+} + x\text{L}$ and equilibrium constant of this is known as instability constant. Which is a measure of stability. The stability of complex depends on EAN of central atom. Charge on metal ion, basic nature of ligand, Chelation

25. Which one of the following does not follow EAN rule.

- a) $\text{Fe}(\text{CO})_5$ b) $\text{V}(\text{CO})_6$ c) $\text{K}_4[\text{Fe}(\text{CN})_6]$ d) $\text{Mn}_2(\text{CO})_{10}$

26. Which complex is most stable where K_a is instability constant

- a) $[\text{Cu}(\text{CN})_2]^-$ $k_a = 1 \times 10^{-16}$ b) $[\text{Fe}(\text{CN})_6]^{4-}$ $k_a = 1 \times 10^{-37}$
c) $[\text{Fe}(\text{CN})_6]^{3-}$ $k_a = 1 \times 10^{-44}$ d) $[\text{Ag}(\text{CN})_2]^-$ $k_a = 1 \times 10^{-20}$

Matrix Matching Type Questions

27. Match Column-I (Co-ordination compounds) with Column-II (Type of isomerism) and select the correct answer using the codes given below the Columns :

Column - I

- A) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$
B) $\text{Cis-}[\text{Co}(\text{en})_2\text{Cl}_2]$
C) $[\text{Co}(\text{en})_2(\text{NO}_2)\text{Cl}]\text{SCN}$
D) $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$

Column - II

- p) Optical isomerism
q) Ionization isomerism
r) Co-ordination isomerism
s) Geometrical isomerism

28. **Column - I**

- A) Ferrocene
B) $\text{Mn}_2(\text{CO})_{10}$
C) Vitamin B_{12}
D) Heame

Column - II

- p) Iron
q) Cobalt
r) Metal-Metal bonding
s) Sandwich structure

29. Column-I

(Pairs of complex compound)

- A) $[\text{Ni}(\text{CO})_4]$ and $\text{K}_2[\text{Ni}(\text{CN})_4]$
 B) $[\text{Cu}(\text{NH}_3)_4]$ and $\text{K}_3[\text{Cu}(\text{CN})_4]$
 C) $\text{K}_2[\text{NiCl}_4]$ and $\text{K}_4[\text{Ni}(\text{CN})_4]$
 D) $\text{K}_2[\text{NiCl}_4]$ and $\text{K}_2[\text{PtCl}_4]$

30. Column-I

- A) $\text{CoCl}_3 \cdot 3\text{NH}_3$
 B) $\text{Co}(\text{NO}_2)_3 \cdot 2\text{KNO}_2 \cdot \text{NH}_3$
 C) $\text{Co}(\text{NO}_2)_3 \cdot 3\text{KNO}_2$
 D) $\text{Co}(\text{NO}_2)_3 \cdot \text{KNO}_2 \cdot 2\text{NH}_3$

Column-II

(Property which is different in given pair)

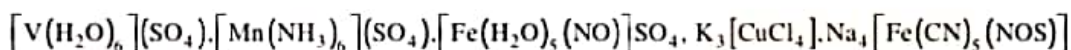
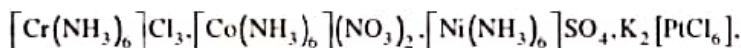
- p) Magnetic moment
 q) oxidation no. of central metal
 r) Geometry
 s) EAN of central atom

Column-II

- p) 600
 q) 0
 r) 900
 s) 1650

Integer Type Questions

31. Find out maximum number of geometrical isomers theoretically possible for an octahedral complex of formula $[\text{Pt}(\text{en})(\text{NO}_2)(\text{Cl})(\text{Br})(\text{I})]$
32. The ratio of terminal to bridged CO group in $[\text{Co}_2(\text{CO})_8]$ is
33. In the reaction $[\text{CoCl}_2(\text{NH}_3)_4]^- + \text{Cl}^- \rightarrow [\text{CoCl}_3(\text{NH}_3)_3] + \text{NH}_3$. If the reactant complex is cis, how many isomers of product complex are formed?
34. FeSO_4 reacts with excess KCN solution to form a complex then no. of unpaired d-e⁻ in central atom of that complex is
35. For the complex MC_2 stepwise formation constants for $\text{M} + \text{L} \rightleftharpoons \text{ML}$ and $\text{ML} + \text{L} \rightleftharpoons \text{ML}_2$ are 4 and 3 overall stability constant is xy. The value of x + y is
36. How many π -bonds are present in ferrocene?
37. Number of possible isomers for $[\text{Pt}(\text{NO}_2)(\text{Py})(\text{NH}_3)\text{Br}]$?
38. Find out total number of paramagnetic complexes which are inner orbital complexes:



KEY SHEET

- 1) bd 2) cd 3) acd 4) ad 5) ab 6) ab 7) abd 8) acd 9) b 10) ac
 11) abcd 12) ad 13) bd 14) ac 15) ab 16) bc 17) bd 18) abd 19) abd 20) bcd
 21) a 22) b 23) b 24) a 25) b 26) c 27) A-s; B-ps; C-pqs; D-r
 28) A-ps; B-r; C-q; D-p 29) A-qrs; B-pqrs; C-pqs; D-prs 30) A-q; B-r; C-s; D-p
 31) 6 32) 3 33) 2 34) 0 35) 3 36) 6 37) 6 38) 3